Rational Expressions

ClipSAT Printable Practice Suite (clipsat.org) • Topic 10.01

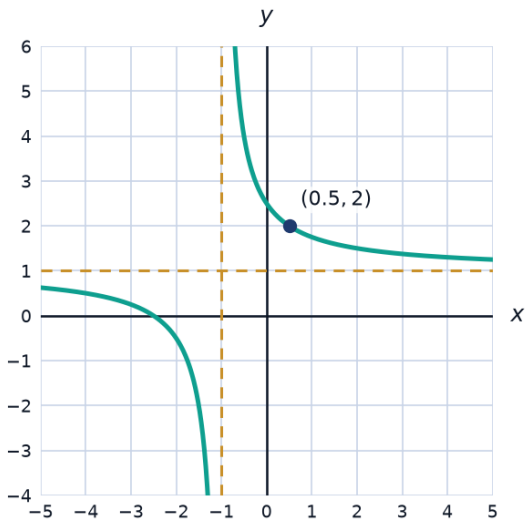
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Key features of quadratic functions

Consider the quadratic function $f(x) = x^2 - 6x + 8$. What are the roots of the equation $f(x) = 0$?

2 [Foundational] • Target: Solving quadratic equations by factoring

Consider the polynomial $P(x) = x^3 - 6x^2 + kx + 12$. If $(x - 3)$ is a factor of $P(x)$, and the sum of the roots of $P(x) = 0$ is S, find the value of k. Additionally, determine the sum of the roots S given that the roots are α, β, γ.



3 [Foundational] • Target: Solving quadratic equations using square roots

The rational expression $\frac{2x^2 + ax - 7}{x + 4}$ is equivalent to $2x + b + \frac{5}{x + 4}$ for all $x \neq -4$, where a and b are constants. What is the value of $a + b$?

4 [Foundational] • Target: Solve quadratic equations using the quadratic formula

The quadratic function f is defined by $f(x) = (x + 3)^2 - 5$. What are the coordinates of the vertex of the graph of $y = f(x)$ in the xy-plane?

5 [Foundational] • Target: Graphing quadratic functions

The quadratic function $f(x) = ax^2 + bx + c$ has a vertex at the point $(2, -5)$. If the graph of $f(x)$ passes through the point $(4, 3)$, find the value of a.

6 [Foundational] • Target: Transformations of quadratics

Factor the quadratic expression: $x^2 + 5x + 6$.

© ClipSAT — clipsat.org | All Rights Reserved

7 [Foundational] • Target: Key features of quadratic functions

Consider the rational expression: $\frac{x^2-4}{x-2}$. If $xeq2$, what is the simplified form of this expression?

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Solving quadratic equations by factoring

Given the polynomial $P(x) = ax^3 - 2x^2 + bx + 5$, if $P(x)$ is divisible by $(x - 1)$ and the remainder when $P(x)$ is divided by $(x + 2)$ is -9, find the value of $a + b$.

9 [Intermediate] • Target: Solving quadratic equations using square roots

If the expression $\frac{\sqrt[4]{x^3 \cdot \sqrt{x^k}}}{x^{-\frac{1}{4}}}$ is equivalent to x^2 for all $x > 0$, what is the value of the constant k?

10 [Intermediate] • Target: Solve quadratic equations using the quadratic formula

Triangle ABC is mapped to triangle $A'B'C'$ on the coordinate plane as shown below. Which coordinate rule represents this transformation?

11 [Intermediate] • Target: Graphing quadratic functions

Let $f(x) = \sqrt{x^2 - 4x + 4}$ and $g(x) = \sqrt[3]{x^3 + 6x^2 + 12x + 8}$. If $h(x) = f(x) \cdot g(x)$, find the value of $h(-3)$.

12 [Intermediate] • Target: Transformations of quadratics

Consider the function defined by $f(x) = \sqrt{ax + b}$. If the graph of $f(x)$ passes through the points $(1, 3)$ and $(5, 5)$, determine the value of a.

13 [Intermediate] • Target: Key features of quadratic functions

In the figure below, triangle ABC is a right triangle with right angle at C, and altitude CD is drawn to hypotenuse AB. The length of altitude CD is 6, $AD = x$, and $DB = x + 5$. What is the length of hypotenuse AB?

14 [Intermediate] • Target: Solving quadratic equations by factoring

The graph of the function $y = a \cdot b^x$, where $ae0$ and $b > 0, be1$, passes through the points $(1, 6)$ and $(3, 54)$. Determine the exact value of a and the exact value of b.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

© ClipSAT — clipsat.org | All Rights Reserved

15 [Challenging] • Target: Solving quadratic equations using square roots • Cross-Lesson: 5.01 Exponential & Logarithmic Functions

Consider a line that passes through the point $(2, 5)$ and has a slope of -3. If the equation of this line is written in the form $y = mx + c$, and it is also known that the point $(k, -7)$ lies on this line, determine the value of k.

16 [Challenging] • Target: Solve quadratic equations using the quadratic formula • Cross-Lesson: 6.01 Systems & Inequalities

The graph of the function $f(x) = a \cdot b^x$ passes through the points $(0, 3)$ and $(2, 75)$. Determine the value of a.

17 [Challenging] • Target: Graphing quadratic functions • Cross-Lesson: 7.01 Sequences & Series

Consider a line passing through the points $A(2, k)$ and $B(6, 10)$. If the slope of the line segment AB is m and $m = \frac{1}{2}$, find the value of k.

18 [Challenging] • Target: Transformations of quadratics • Cross-Lesson: 8.01 Factoring Polynomials

The graph of the function $f(x) = \sqrt{ax + b}$ passes through the points $(1, 3)$ and $(5, 5)$. Find the value of a and the value of b.

19 [Challenging] • Target: Key features of quadratic functions • Cross-Lesson: 9.01 Radicals & Exponents

The graph of the exponential function $f(x) = a \cdot b^x + c$ passes through the points $(0, 3)$ and $(1, 7)$. If the horizontal asymptote of the graph is $y = 1$, find the value of a and b.

20 [Challenging] • Target: Solving quadratic equations by factoring • Cross-Lesson: 1.01 Linear Equations & Lines

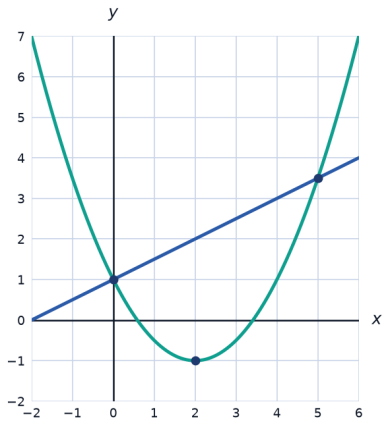
Let $P(x)$ be a polynomial such that $P(x^2 + 1) = x^4 + 3x^2 + 2$. If $P(x)$ can be expressed in the form $ax^2 + bx + c$, find the value of b.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

© ClipSAT — clipsat.org | All Rights Reserved

21 [Foundational] • Target: Key features of quadratic functions

For the model $f(x) = 2x + 6$ in key features of quadratic functions, evaluate $f(2)$.



a. 10

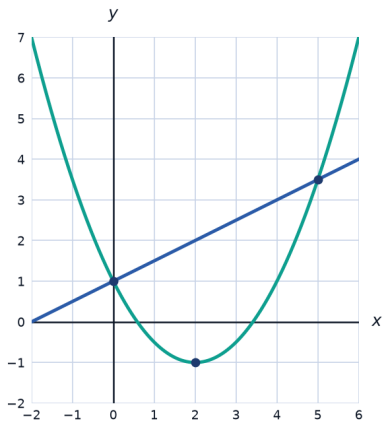
b. 12

c. 4

d. 14

22 [Foundational] • Target: Key features of quadratic functions

Find the positive solutions to the quadratic equation $x^2 - 6x + 8 = 0$.



a. $x = 2$ and $x = 5$

b. $x = 2$ and $x = 4$

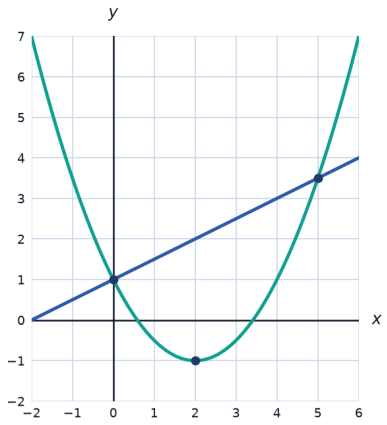
c. $x = -2$ and $x = -4$

d. $x = 6$ and $x = 8$

© ClipSAT — clipsat.org | All Rights Reserved

23 [Foundational] • Target: Key features of quadratic functions

For the model $f(x) = 2x + 6$ in key features of quadratic functions, evaluate $f(5)$.



a. 16

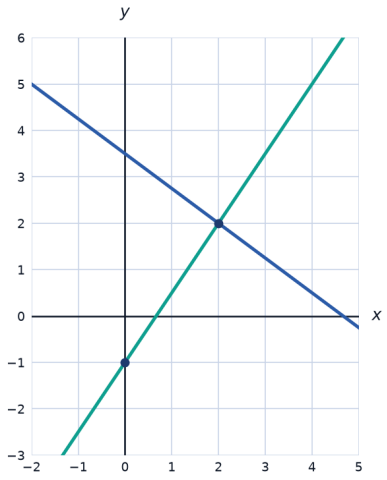
b. 18

c. 10

d. 20

24 [Foundational] • Target: Solving quadratic equations by factoring

For the model $f(x) = 3x + 7$ in solving quadratic equations by factoring, evaluate $f(2)$.



a. 13

b. 16

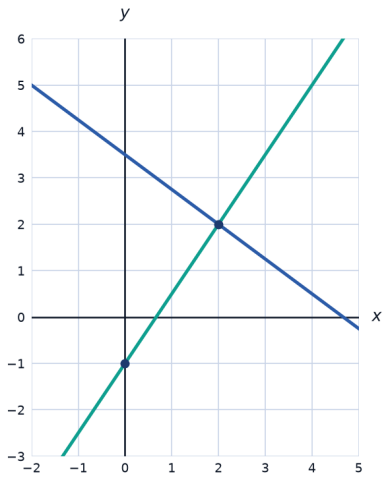
c. 6

d. 17

© ClipSAT — clipsat.org | All Rights Reserved

25 [Foundational] • Target: Solving quadratic equations by factoring

Find the positive solutions to the quadratic equation $x^2 - 7x + 10 = 0$.



a. $x = 2$ and $x = 6$

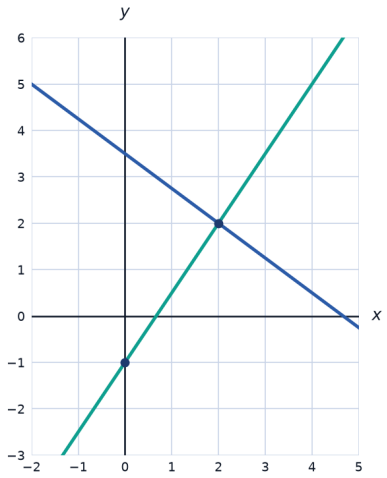
b. $x = 2$ and $x = 5$

c. $x = -2$ and $x = -5$

d. $x = 7$ and $x = 10$

26 [Foundational] • Target: Solving quadratic equations by factoring

For the model $f(x) = 3x + 7$ in solving quadratic equations by factoring, evaluate $f(5)$.



a. 22

b. 25

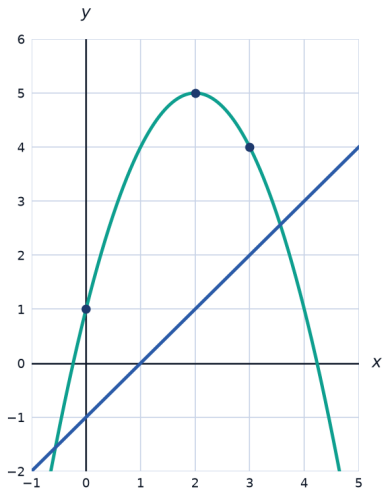
c. 15

d. 26

© ClipSAT — clipsat.org | All Rights Reserved

27 [Foundational] • Target: Solving quadratic equations using square roots

For the model $f(x) = 4x + 8$ in solving quadratic equations using square roots, evaluate $f(2)$.



a. 16

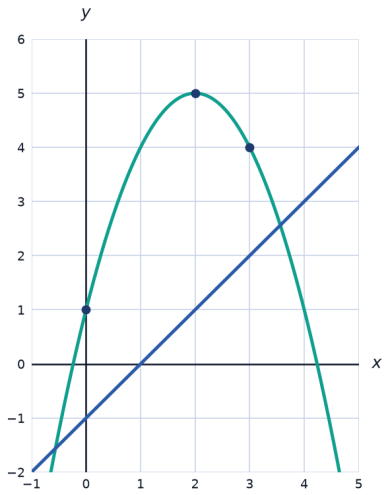
b. 20

c. 8

d. 20

28 [Foundational] • Target: Solving quadratic equations using square roots

Find the positive solutions to the quadratic equation $x^2 - 8x + 12 = 0$.



a. $x = 2$ and $x = 7$

b. $x = 2$ and $x = 6$

c. $x = -2$ and $x = -6$

d. $x = 8$ and $x = 12$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

© ClipSAT — clipsat.org | All Rights Reserved

29 [Intermediate] • Target: Key features of quadratic functions

Find the positive solutions to the quadratic equation $x^2 - 4x + 3 = 0$.

a. $x = 1$ and $x = 4$

b. $x = 1$ and $x = 3$

c. $x = -1$ and $x = -3$

d. $x = 4$ and $x = 3$

30 [Intermediate] • Target: Key features of quadratic functions

A data set recorded for key features of quadratic functions consists of the four values $\{18, 22, 26, 32\}$. What is the arithmetic mean?

a. 22.5

b. 23.5

c. 24.5

d. 26.5

31 [Intermediate] • Target: Key features of quadratic functions

For the model $f(x) = 2x + 6$ in key features of quadratic functions, evaluate $f(4)$.

a. 14

b. 16

c. 8

d. 18

32 [Intermediate] • Target: Key features of quadratic functions

Find the positive solutions to the quadratic equation $x^2 - 10x + 24 = 0$.

a. $x = 4$ and $x = 7$

b. $x = 4$ and $x = 6$

c. $x = -4$ and $x = -6$

d. $x = 10$ and $x = 24$

33 [Intermediate] • Target: Solving quadratic equations by factoring

A data set recorded for solving quadratic equations by factoring consists of the four values $\{10, 14, 18, 24\}$. What is the arithmetic mean?

a. 14.5

b. 15.5

c. 16.5

d. 18.5

© ClipSAT — clipsat.org | All Rights Reserved

34 [Intermediate] • Target: Solving quadratic equations by factoring

For the model $f(x) = 3x + 7$ in solving quadratic equations by factoring, evaluate $f(3)$.

a. 16

b. 19

c. 9

d. 20

35 [Intermediate] • Target: Solving quadratic equations by factoring

Find the positive solutions to the quadratic equation $x^2 - 9x + 18 = 0$.

a. $x = 3$ and $x = 7$

b. $x = 3$ and $x = 6$

c. $x = -3$ and $x = -6$

d. $x = 9$ and $x = 18$

36 [Intermediate] • Target: Solving quadratic equations using square roots

Find the positive solutions to the quadratic equation $x^2 - 6x + 5 = 0$.

a. $x = 1$ and $x = 6$

b. $x = 1$ and $x = 5$

c. $x = -1$ and $x = -5$

d. $x = 6$ and $x = 5$

37 [Intermediate] • Target: Solving quadratic equations using square roots

For the model $f(x) = 4x + 8$ in solving quadratic equations using square roots, evaluate $f(3)$.

a. 20

b. 24

c. 12

d. 24

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Solving quadratic equations using square roots • Cross-Lesson: 8.01 Factoring Polynomials

For the model $f(x) = 4x + 8$ in solving quadratic equations using square roots, evaluate $f(4)$.

a. 24

b. 28

c. 16

d. 28

© ClipSAT — clipsat.org | All Rights Reserved

39 [Challenging] • Target: Solving quadratic equations using square roots • Cross-Lesson: 9.01 Radicals & Exponents

Find the positive solutions to the quadratic equation $x^2 - 12x + 32 = 0$.

a. $x = 4$ and $x = 9$

b. $x = 4$ and $x = 8$

c. $x = -4$ and $x = -8$

d. $x = 12$ and $x = 32$

40 [Challenging] • Target: Solve quadratic equations using the quadratic formula • Cross-Lesson: 1.01 Linear Equations & Lines

A data set recorded for solve quadratic equations using the quadratic formula consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean?

a. 16.5

b. 17.5

c. 18.5

d. 20.5

41 [Challenging] • Target: Solve quadratic equations using the quadratic formula • Cross-Lesson: 11.01 Statistics & Data Analysis

For the model $f(x) = 5x + 9$ in solve quadratic equations using the quadratic formula, evaluate $f(4)$.

a. 29

b. 34

c. 20

d. 33

42 [Challenging] • Target: Solve quadratic equations using the quadratic formula • Cross-Lesson: 2.01 Quadratic Functions

Find the positive solutions to the quadratic equation $x^2 - 11x + 24 = 0$.

a. $x = 3$ and $x = 9$

b. $x = 3$ and $x = 8$

c. $x = -3$ and $x = -8$

d. $x = 11$ and $x = 24$

43 [Challenging] • Target: Graphing quadratic functions • Cross-Lesson: 3.01 Exponents & Polynomials

A data set recorded for graphing quadratic functions consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean?

a. 17.5

b. 18.5

c. 19.5

d. 21.5

© ClipSAT — clipsat.org | All Rights Reserved

44 [Challenging] • Target: Graphing quadratic functions • Cross-Lesson: 4.01 Functions & Transformations

For the model $f(x) = 6x + 3$ in graphing quadratic functions, evaluate $f(4)$.

a. 27

b. 33

c. 24

d. 31

45 [Challenging] • Target: Transformations of quadratics • Cross-Lesson: 5.01 Exponential & Logarithmic Functions

For the model $f(x) = 2x + 4$ in transformations of quadratics, evaluate $f(3)$.

a. 10

b. 12

c. 6

d. 14

© ClipSAT — clipsat.org | All Rights Reserved